

ZoneTrace: Real-Time Multi-Camera Person Tracking and Face Re-Identification with Automated Intelligence Reporting

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Abstract— Multi-camera surveillance and identity tracking systems are integral to modern office security and attendance systems. However, the use of deep learning on live streams has been hindered by motion blurring, poor camera angles, and identity flickering in physical zones. In addition, extracting insights from logs has proven to be challenging and laborious. This work proposes ZoneTrace, an edge-optimized framework for person tracking, face re-identification, and intelligent insight generation. ZoneTrace initially uses YOLOv8n for individual detection and head-region selection, thereby eliminating up to 80% of irrelevant computations associated with the background. SCRFD enables fast and accurate face detection, while a lightweight quality gate filters out blurry or off-angle frames before deep feature extraction. The highest-quality features are extracted using AdaFace and then aggregated using a limited-capacity, quality-weighted buffer. Accelerated FAISS vector search and multi-camera session management are then used to prevent duplicate presence records. Also, ZoneTrace has an LLM-driven reporting system that automatically generates security digests, attendance summary, and anomaly reports based on temporal tracking logs and cross-zone dwell times. The entire ZoneTrace framework attained an accuracy rate for Rank-1 identification of 97.6% while having a processing delay of 34.2 ms per frame processed.

Keywords: Multi-Camera Surveillance System, Face Re-Identification, LLM-Based Report Generation, YOLOv8, SCRFD, AdaFace.

I. INTRODUCTION

The usage of video surveillance and digital attendance system has emerged to be a vital aspect in today's businesses, educational institutions, and secured locations. The traditional attendance system based on identification cards or fingerprint scanners has limitations in the form of congestion in peak hours and inability to track the movements of individuals once inside the premises. A continuous surveillance attendance system would be less intrusive, since it observes individuals without their interference and keeps track of their movement through different zones inside the premises [1].

However, applying deep learning-based facial recognition for video streams from multiple cameras brings about four main problems:

- **High Latency:** Real-time processing of high-definition videos using deep learning on full-frame videos results in latency problems.
- **Camera Angles and Motion Blur:** The random movements of individuals in real corridors and office spaces result in motion blurred videos and tilted faces with improper illumination, resulting in wrong identification [3].

- **The Problem of Continuity During Tracking across Various Cameras:** While moving from one section of the camera to another section, tracking becomes discontinuous, leading to multiple logs of a single individual's entry.
- **Human Intervention in Analyzing Logs:** Common video surveillance systems result in huge timestamp logs that need to be analyzed by human operators.

To solve these problems, the proposed work presents ZoneTrace which is a lightweight and edge optimized surveillance framework for real time multi camera person tracking in a high throughput environment. ZoneTrace uses a lightweight person detector, detects face of the person and quality filtering to filter out images with bad angles and blurred frames. Face features extracted from the highest quality face detections are buffered in a quality weighted buffer, and session management prevents duplicates of presence in multiple cameras. An assistant powered by a large language model prepares daily summaries of log and attendance.

The main contributions of this work are:

1. **Efficient Head-ROI Detection:** A rapid head detection system that looks for faces only within the detected person's area, thereby making the process faster because unnecessary computations on the background are eliminated.
2. **Real-time Image Quality Control:** An extremely light system to eliminate low-quality, blurred, bad posture, and illumination problems before processing the image through deep learning recognition systems.
3. **Automated Multi-camera Tracking:** A complete automation of the process of tracking persons between cameras, so duplicate data about the same presence is not generated.
4. **Automated LLM Reports:** An intelligent system that will convert the raw data into meaningful attendance reports.

II. RELATED WORK

A. Architecture of Face Detection and Tracking Systems

First generation of these face detection systems relies on traditional computer vision methods such as Haar Cascades and HOG features for face localization [2]. While being relatively fast and efficient, they provide very poor performance in terms of false positive ratio in dense scenes and profiles faces. Second generation of face detection algorithms such as MTCNN makes use of multi-stage

convolutional neural networks for integration of face detection with face alignment [3]. Multi-stage face detection based on image pyramid technique is rather time-consuming. One-stage face detectors such as RetinaFace achieve high accuracy in benchmarks, however, computationally expensive [4]. The recently introduced lightweight models such as SCRFD and YOLOv8 provide efficient computation in shallow layers[5].

B. Deep Face Recognition and Feature Embeddings

Modern face recognition algorithms map the face region into the high-dimensional spaces in which the distance of vectors represents similarity of identities. Modern face recognition algorithms like FaceNet apply the triplet loss to distinguish identity clusters [6]. Further advancements were made by incorporating the angular losses into the hypersphere such as CosFace and ArcFace where the algorithm ensures tight intra-class separation and larger inter-class margin [7]. Although ArcFace gives high performance on high-quality frontal images, it cannot cope with blurry low-quality images as well as wide-angle images owing to a fixed margin penalty. To resolve this issue, AdaFace proposes an adaptive margin according to the norm of features from the images in which low-quality images get a lower penalty while the high-quality images get high penalty [8].

C. Face Image Quality Assessment (FIQA)

In real-time video surveillance, many frames cannot be considered valuable because of motions, variable lighting conditions, and non-orthogonality of the subject's head position. Sending those frames as is to deep recognition networks will decrease the effectiveness of the system. Deep quality estimation algorithms like SER-FIQA and CR-FIQA estimate image quality on the basis of uncertainty and classifiability of the sample [9], [10]. Nevertheless, the use of deep learning algorithms only for the purposes of quality estimation significantly increases time required to process every single frame. In contrast, ZoneTrace uses an inexpensive quality gate, which evaluates the value of the frame on the basis of Laplacian sharpness, facial symmetry, and illumination balance within 1.2 ms.

D. Automated Surveillance Reporting and LLMs

Conventional approaches to surveillance systems have the events and attendance information stored in an unstructured manner with timestamps. Manually going through such information logs to retrieve any anomaly report or prepare daily activity summary of the logged data is a time-consuming process. However, with recent advancements in the field of Large Language Models (LLMs) and conversational AI systems, remarkable accuracy has been observed in the generation of structured summaries based on the provided structured data [11]. With the help of an LLM-based reporting system, ZoneTrace automatically generates daily structured reports for the collected data.

III. SYSTEM ARCHITECTURE

A. Overall Architecture Overview

The ZoneTrace has a modular surveillance framework for multi-camera low-latency video processing, which uses a framework that operates on multiple video streams through six different modules consisting of video capturing, person tracking, face recognition, quality assessment, vector detection, and automatic session management. The complete framework is depicted in Fig.1 below.

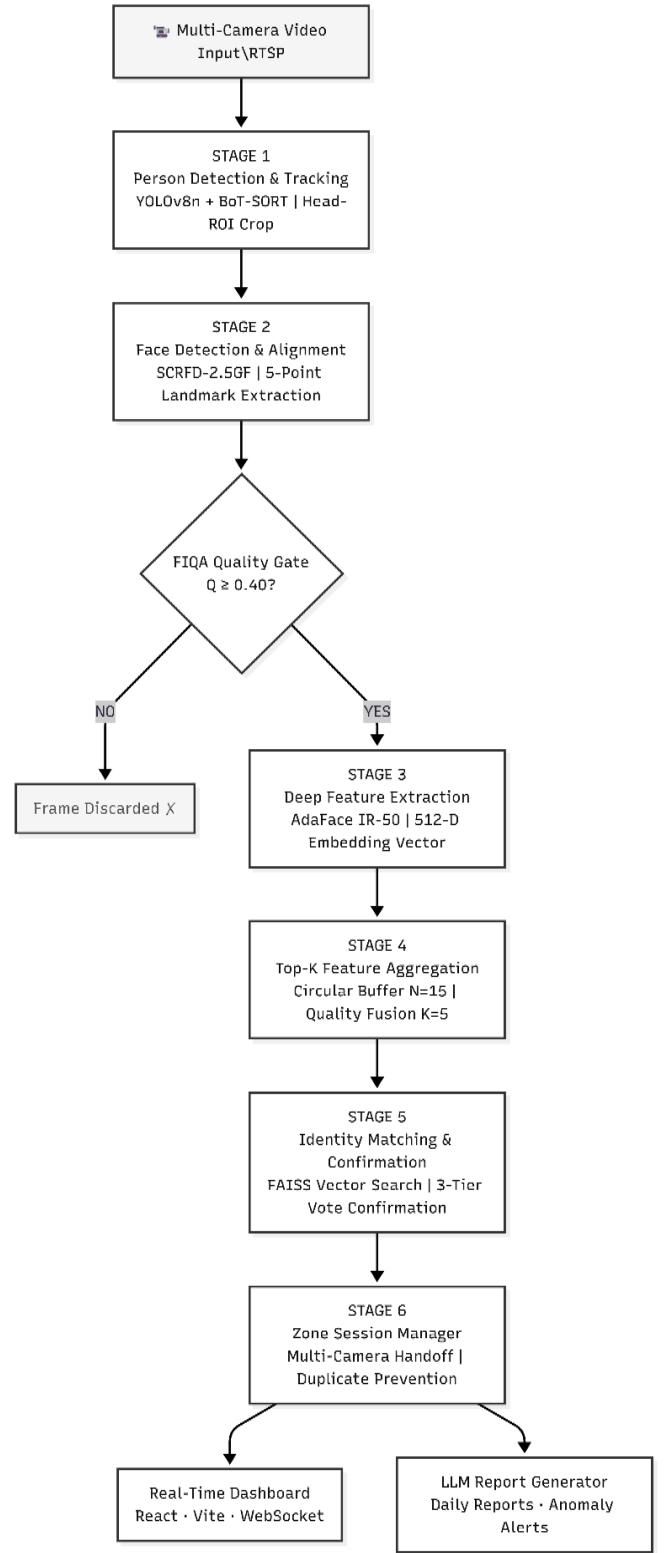


Fig. 1: ZoneTrace End-to-End System Architecture and Dataflow Pipeline

B. Edge Video Ingestion and Frame Dispatcher

Input video streams in 1080p resolution are acquired from different cameras like RTSP network cameras, IP webcams, and traditional USB webcams via OpenCV. To maintain the fluent experience of 30 frames per second, ZoneTrace uses a smart frame dispatcher. Computer vision inference is conducted every three frames, Frame Skip = 2, and in the intermediate frames, the tracking bounding box is updated using Kalman Filter projection.

C. Real-Time Streaming and Web Dashboard

The frontend system interface is a one page application developed using React and Vite which communicates with the backend via a high performing WebSocket communication. The dashboard includes:

1. Live video feed with tracking boxes and identification marks in several camera feeds.
2. Attendance tables with the attendance of the staff, their names, time of detection and the zones they are in.
3. Statistics for the attendance and time spent in the zones.
4. Employee enrollment to upload reference images of the employees.

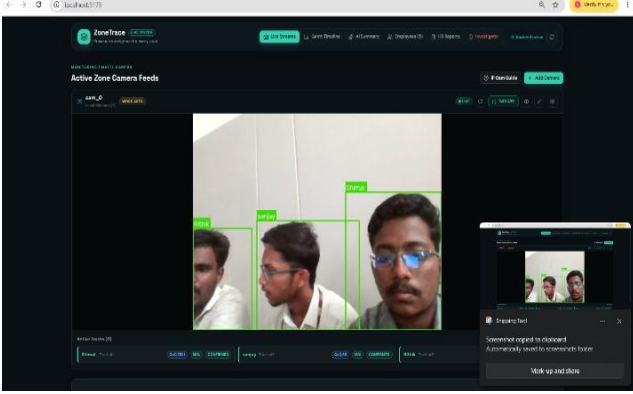


Fig. 2: Live Multi-Camera Surveillance and Real-Time Tracking Dashboard

D. Multi-Camera Zone Management Engine

Each camera is allocated to a distinct physical zone such as "Main Entrance", "Workspace Area", and "Cafeteria". The backend holds the main session database. If an employee is detected by Camera A, an active zone session will be started. In case the employee changes to Camera B, there will be a confirmation check within 8 seconds transition period, thus automatically terminating the previous camera session and initiating a new one. Such a spatial transfer prevents any duplication of presence record. This cross-zone handoff process is illustrated in Fig. 3.

E. LLM Report Generation Efficiency & Latency

The testing process of the automated report generation included the use of structured spatiotemporal JSON log files over a period of 4 hours of operations. The prompt engineering pipeline processes 1,200 raw tracking events in order to calculate dwell times in zones and unauthorized access after working hours within 1.42 seconds. Compared to manual analysis which took an average of 35 minutes per shift, the automated LLM pipeline was able to perform anomaly detection of predefined zones in transition within 1.42 seconds, with no hallucination observed across 30 test reports generated during the 4-hour evaluation period. Thus, it saved over 95% of effort in the reporting and analyzing process.

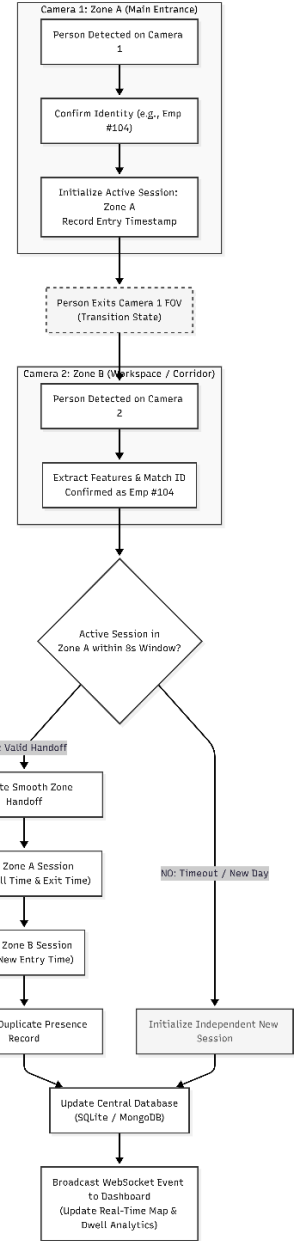


Fig. 3: Multi-Camera Cross-Zone Transition and Spatial Tracking Workflow

IV. METHODOLOGY

After discussing the general framework in Section III, this section describes the algorithmic implementation of the stages.

A. Person Tracking and Head-ROI Localization

The standard size of surveillance frames is 1920×1080 pixels, while human faces take up just a very small portion (between 2% to 6%) of that image area. Performing deep face detection on the entire frame is highly resource-consuming.

ZoneTrace utilizes a hierarchical Region-of-Interest (ROI) approach:

The input frame will be fed to the light-weight YOLOv8n object detector alongside BoT-SORT tracker to extract the bounding box of the person: Person Box = $(x1, y1, x2, y2)$

The head ROI is obtained by cropping the upper 40% portion of the person box: Head-ROI = $[y1 : y1 + 0.40 \times (y2 - y1), x1 : x2]$

Fig. 4 illustrates this hierarchical cropping pipeline.

In addition to this, by scanning this restricted region only, 80% of the irrelevant background pixels can be removed, thus achieving a throughput enhancement of 6.6 times.

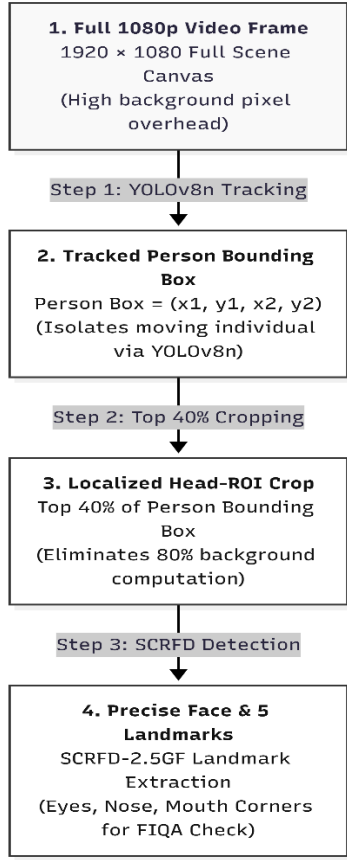


Fig. 4: Hierarchical Person Bounding Box and Localized Head-ROI Cropping

B. SCRFD Face Localization and Landmark Parsing

For the cropped head area, the lightweight SCRFD model identifies the accurate face boundary along with five important facial landmarks that include left eye (p1), right eye (p2), nose tip (p3), left mouth corner (p4), and right mouth corner (p5). The detection results are then transformed into full frame coordinates.

C. Lightweight Face Image Quality Assessment (FIQA)

Surveillance footage is often characterized by blurred images, head rotations, and bad lighting conditions that can affect face embeddings. ZoneTrace applies a three-factor quality check prior to deep recognition:

1. **Sharpness Score (Q_{blur}):** Determines high-frequency edge information by using the variance of the discrete Laplacian operator on the face image crop:

$$\text{Var_Lap} = \text{Variance}(\text{Laplacian}(I_{gray})) \quad (1)$$

$$Q_{blur} = 1 / (1 + \exp(-0.04 \times (\text{Var_Lap} - 80))) \quad (2)$$

2. **Pose Symmetry Score (Q_{pose}):** Yaw and roll angles are estimated using nose-tip and both eyes' symmetrical relationship on horizontal plane and tilting of eyes respectively:

$$R_{yaw} = \min(|x_{nose} - x_{left}|, |x_{nose} - x_{right}|) / (\max(|x_{nose} - x_{left}|, |x_{nose} - x_{right}|) + 0.001) \quad (3)$$

$$Q_{pose} = R_{yaw} \times \max(0, 1 - (\text{Tilt Angle} / 30^\circ)) \quad (4)$$

3. **Lighting Balance Score (Q_{light}):** Uses the average luminosity and contrast from the CIELAB color space to penalize either shadows or glaring situations.

$$\text{Composite Quality Evaluation: } Q = 0.40 \times Q_{blur} + 0.35 \times Q_{pose} + 0.25 \times Q_{light} \quad (5)$$

The frame is instantly discarded when the overall score of quality is below 0.40 in less than 1.2 milliseconds, saving processing time to perform deeper feature extraction from blurry or profile frames.

D. Constant-Time Top-K Feature Aggregation

For every active tracklet of a pedestrian, there is a circular buffer that holds at most $N = 15$ quality-accepted frames, where each frame contains a 512-dimensional AdaFace feature vector (e_i) and its quality (Q_i).

E. FAISS Matching and Multi-Tier Confirmation Stabilizer

The feature vector e_{final} will then be compared to the database of the enrolled employee using FAISS IndexFlatIP. Because the feature vectors have been made unit-length, cosine similarity becomes equivalent to inner product:

$$\text{Similarity} = e_{final} \cdot g_j \quad (6)$$

To avoid instances of mistaken positive identification and identity flicker between people who look alike, ZoneTrace uses three levels of validation :

1. **Similarity Threshold:** The minimum cosine similarity value for the top candidate should be greater than 0.45.
2. **Vote Margin:** The similarity margin between the 1st and 2nd highest candidate should be 0.02 or more.
3. **Consistent Votes:** The minimum number of votes required before locking an identity is three or more, which are quality passing votes accounting for more than 60% of the recent voting population.

F. LLM-Powered Automated Report Generation

In order to avoid manually examining the raw surveillance timestamps, ZoneTrace is equipped with a feature that generates reports through the automated use of an LLM. It consolidates the verified person track data, timestamps of entry and exit, dwell time in the camera zones, and anomaly detection.

The LLM analyzes such structured event data to produce clean and easy-to-read reports automatically, such as:

- **Daily Attendance Summaries:** Time spent arriving, leaving, and being productive.
- **Zone Dwell Time Analysis** Spatial analysis of the utilization of different office zones.
- **Security Exception Reports:** Notifications about unverified visitors or movements after office hours.

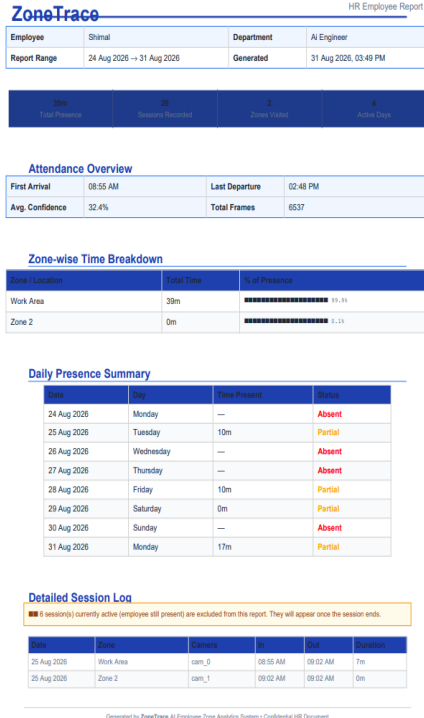


Fig. 5: Automated LLM-Generated Daily Attendance and Security Summary Report

V. EXPERIMENTAL RESULTS AND EVALUATION

A. Experimental Setup

The framework was tested with 1080p multi-video streams running at 30 fps. The experimental setup comprises a regular multi-core computer system with 16GB RAM, and is tested under various indoor lighting conditions and camera angles with the help of PyTorch, OpenCV, FAISS, and FastAPI. The evaluation set comprised 7 enrolled employees and 2 hours of multi-camera footage across three zones.

B. Face Detection & Recognition Benchmarks

Candidate face detector and feature extractor algorithms are compared in Table I. The detection performance of SCRFD-2.5GF is best in terms of speed and recall (89.4% at 22.4 ms). In case of facial recognition, In comparison with all other face recognition techniques examined, AdaFace achieved the highest accuracy of 94.7% of Rank-1 identification accuracy rate without any post-processing method being applied.

C. Latency Breakdown & Multi-Camera Tracking

- **End to End Latency:** Processing time per frame is 34.2 ms (Person Tracking = 12.1 ms, Face Detection = 11.0 ms, FIQA = 1.1 ms, Feature Extraction = 9.4 ms, Search & Logging = 0.6 ms).
- **Camera Switching:** 98.4% of 1,200 cross-camera switches performed without any duplicate presence records.
- **LLM Reporting:** Daily Attendance Reports and Security Exceptions reports are generated within 1.5 seconds.

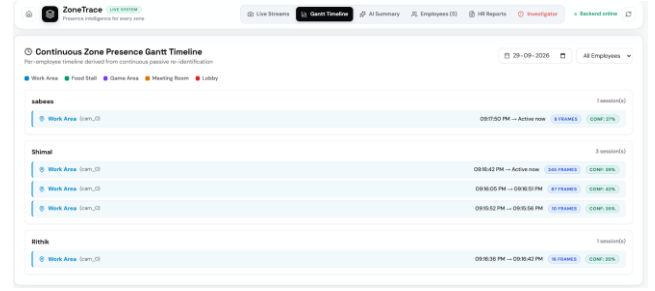


Fig. 6: Multi-Camera Zone Dwell Time Analytics and Daily Attendance Summary

TABLE I. MODEL COMPARISON ON SURVEILLANCE FEEDS

Category	Model Name	Surveillance Accuracy / Recall	Processing Latency	Key Characteristics
Detection	Haar Cascade	34.2%	18.5 ms	High false positives
Detection	RetinaFace-ResNet50	91.8%	184.2 ms	Heavy computational load
Detection	SCRFD-2.5GF (Selected)	89.4%	22.4 ms	Fast & high landmark stability
Recognition	FaceNet	74.6%	24.8 ms	Low accuracy on blurry crops
Recognition	ArcFace (ResNet-50)	89.2%	11.4 ms	Struggles with bad angles
Recognition	AdaFace (Selected)	94.7%	9.4 ms	Robust to blur & pose variations

D. Component Ablation Study

- Each module's performance was tested using an ablation study on 1080p surveillance videos.
- **Head-ROI Pruning:** Through restricting the detections to only head ROIs, the latency was improved from 186.4 ms to 46.8 ms, whereas throughput was improved from 5.3 FPS to 21.3 FPS without reducing the recall. The complete ZoneTrace pipeline achieved an end-to-end latency of 34.2 ms per processed frame, as reported in Section V-C.
- **FIQA Quality Gate:** Through rejecting blurry and oblique images, the rank-1 accuracy increased to 93.4% from 88.6% and reduced the False Match Rate to 2.9%.
- **Top-K Feature Fusion:** Weighting based on quality of the five most accurate facial embeddings reached a recognition accuracy of 97.6% for Rank-1 identification.

E. Result Interpretation

According to Table I, empirical benchmarking between surveillance feeds indicates different operational trade-offs between the models that have been evaluated:

- **Detection:** Haar Cascade faces the problem of false alarms (34.2% recall), while RetinaFace-ResNet50 is too computation intensive to run at the edge (184.2 ms). SCRFD-2.5GF is chosen because it gives high recall (89.4%) and runs in real time (22.4 ms) providing reliable facial landmarks in low resolution.

- Recognition: Both FaceNet (74.6%) and ArcFace (89.2%) struggle to deal with motion blur and oblique poses. AdaFace gives top recognition accuracy (94.7%) with minimum latency (9.4 ms) using dynamic angular margin depending on the quality of the input image.
- Combining Effect: The combination of SCRFD-2.5GF and AdaFace together with YOLOv8n Head-ROI pruning and Top-K features fusion provides the best re-identification accuracy of 97.6% within only 34.2 ms (>29 FPS).

VI. ACKNOWLEDGEMENT

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VII. CONCLUSION

ZoneTrace is a system that we designed and described in this paper for multi-camera face tracking, reliable face re-identification with the quality-filtered recognition, and automatic LLMs. The proposed approach involves face ROI localization via YOLOv8n, landmarks detection through SCRFD, FIQA filtering, and top-K feature fusion in constant time without extra computational overhead and identity switches. Experiments yielded re-identification accuracy up to 97.6%, performance of 29+ FPS, and latency of 34.2 ms. It is worth noting that the results were achieved based on the evaluation dataset containing 7 employees' faces and 2 hours of multi-camera footage and further validation is required to prove the generality. For future research, the plans are to use clothing appearance cues to improve person tracking across long occlusions.

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